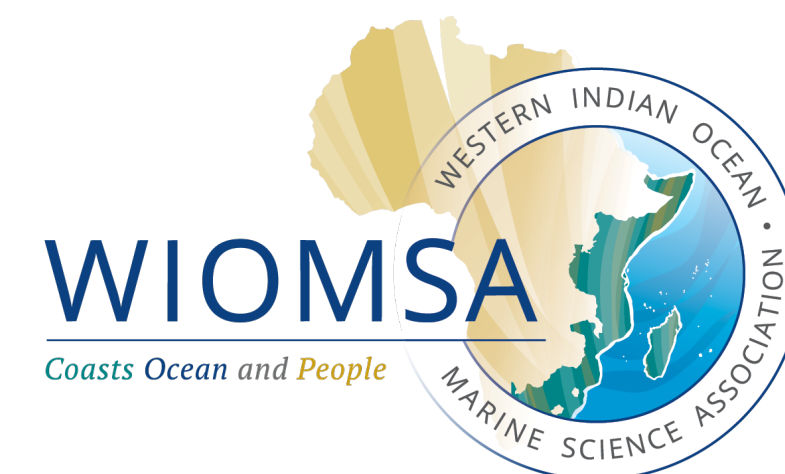

Data Harmonization

A Collaborative Approach to Unified Fisheries Data in East Africa

Lorenzo Longobardi



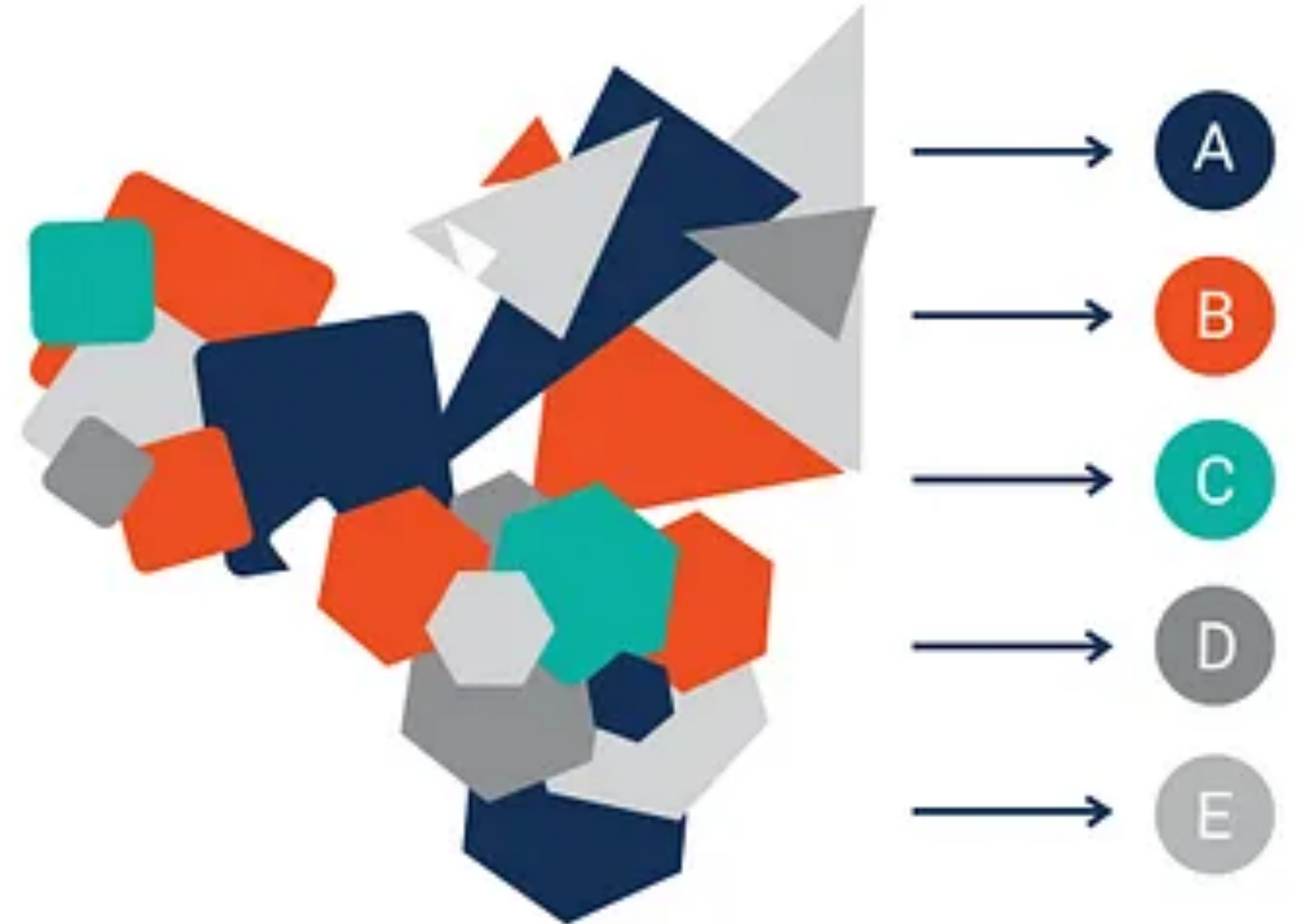
13TH
SCIENTIFIC
SYMPOSIUM



Data collection heterogeneity

One resource, many approaches

1. Information collected
2. Paper vs. Digital forms
3. Terms/variables heterogeneity
4. Species Lists and taxonomic resolution
5. Fishery information



Why Harmonization?

Tangible benefits

- **Stronger visibility & influence** → Harmonized data feeds into national statistics and regional policy platforms (Nairobi Convention, SWIOFC).
- **Better fundraising leverage** → Donors increasingly demand comparable, standardized evidence. Harmonized data positions NGOs for larger, multi-country funding.
- **Faster insights, less burden** → Avoids endless data cleaning/wrangling; automated analytics free staff time for community work and advocacy.
- **Credibility & trust** → Harmonized definitions ensure your data can stand alongside government data in decision-making spaces.
- **Scaling local impact** → What you collect locally contributes directly to regional narratives on food, livelihoods, and climate resilience.

Individual datasets become more powerful when integrated

data:



information:



knowledge:



insight:

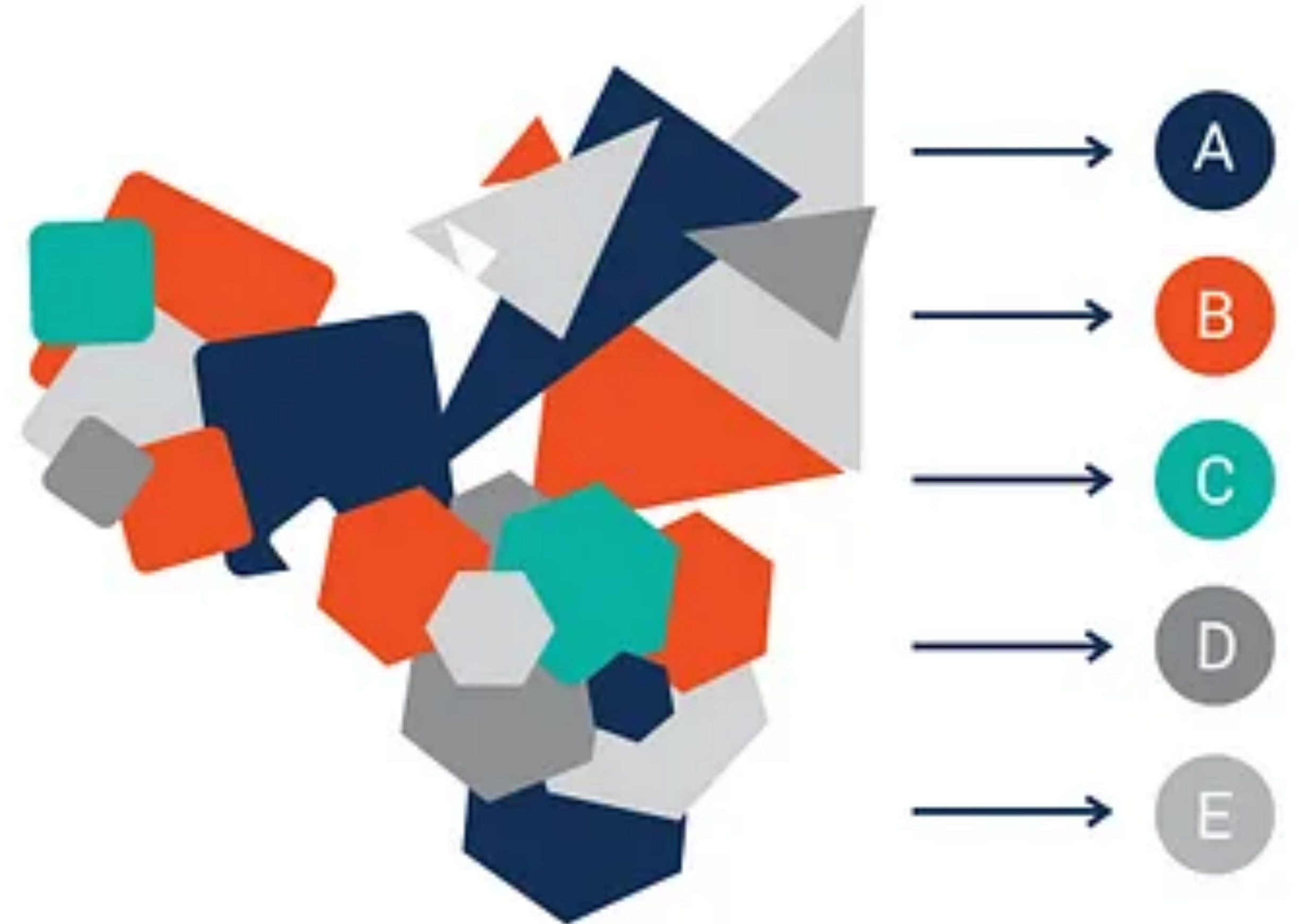


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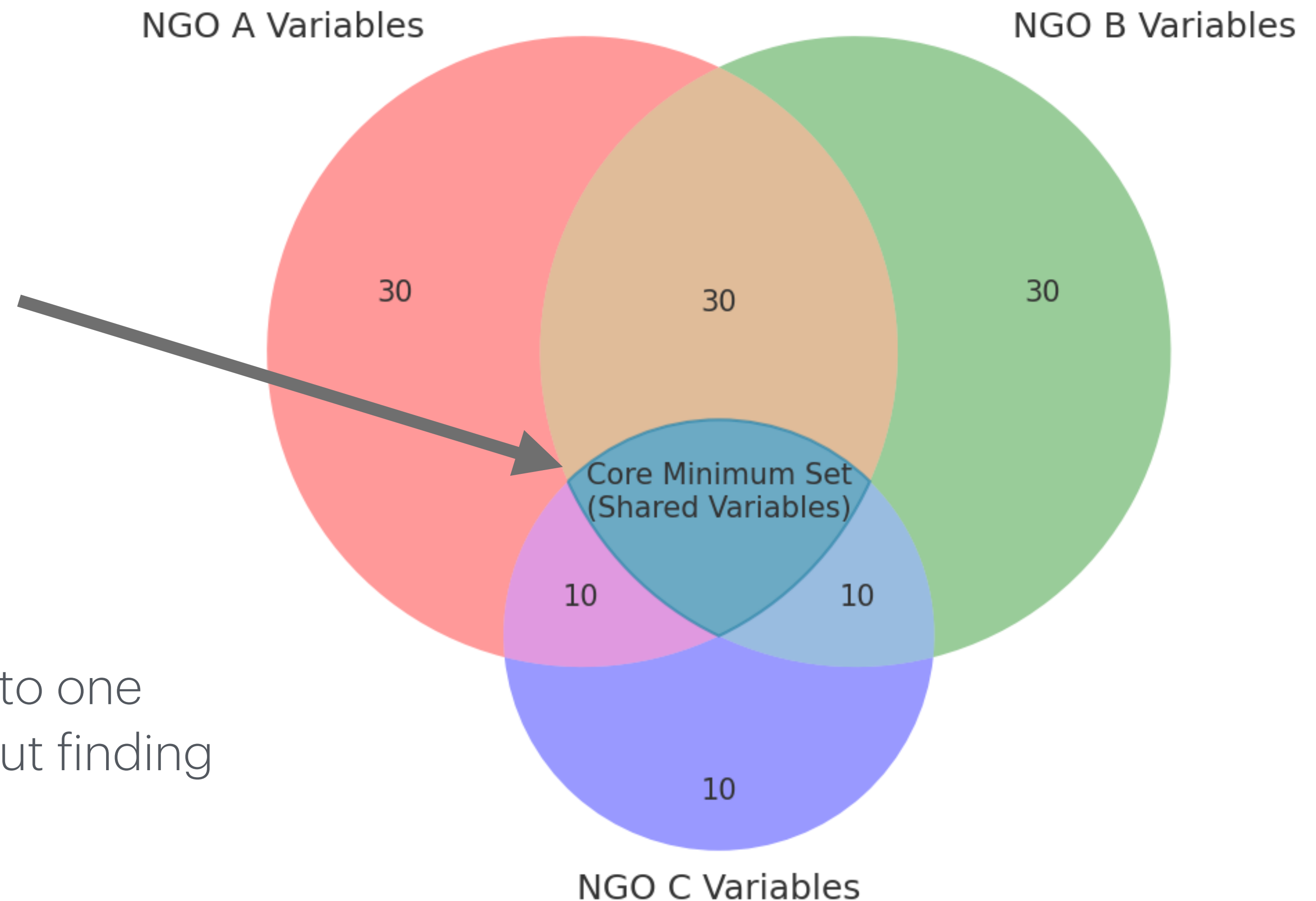


Information collected

Minimum Set of Common Variables

We need a **minimum common denominator** — a set of variables that every dataset must provide, consistently defined.

Harmonization **is not** forcing everyone into one identical form/collection process. It's about finding core overlaps and common definitions



Information collected

CGIAR harmonization strategy example

	Variable name	Data type	Unit	Description	Recommended protocol	Ontology URI
Where and when	Date of Collection	date	YYYY-MM-DD (ISO 8601)	Date when data on fishing activity/catch was recorded	Use ISO 8601 format for full interoperability	
	Landing Site	categorical	site name / ID	Location at which boats land their catch	Each site should have geographical coordinates	http://w3id.org/aqfo/aqfo_00000030
	Vessel / Boat Type	categorical	text / code list	Type of vessel or boat used for fishing		http://w3id.org/aqfo/aqfo_00001013
Effort	Number of fishing trips per time unit	integer	trips per day/week/month	Frequency of fishing trips in a given time unit	Enumerator recall survey or logbook records	http://w3id.org/aqfo/aqfo_00000021
	Gear Type	categorical	text / code list	Fishing gear used during the trip	Use FAO ISSCFG (International Standard Statistical Classification of Fishing Gear) where applicable	http://w3id.org/aqfo/aqfo_00002220
	Number of fishers for fishing trip	integer	persons	Number of crew members actively fishing during the trip		http://w3id.org/aqfo/aqfo_00000022
	Duration of fishing trip (hours)	decimal	hours	Total time spent fishing during a single trip		http://w3id.org/aqfo/aqfo_00002011
Catch	Fish group	categorical	taxonomic group	Taxonomic grouping of catch (e.g., small pelagics, demersals, reef fish)	Use FAO ASFIS list	http://w3id.org/aqfo/AQFO_00015095
	Fish length	decimal	cm	Length of individual fish sampled from the catch	Standard length, fork length, or total length (specify consistently)	http://w3id.org/aqfo/aqfo_00002073
Others	Catch price	decimal	local currency	Price of catch at the landing site		http://w3id.org/aqfo/aqfo_00002202
	Catch habitat	categorical	text (reef, open sea, river, mangrove, etc.)	Habitat in which the fishing activity occurred		http://w3id.org/aqfo/aqfo_00000023

Data collection heterogeneity

One resource, many approaches

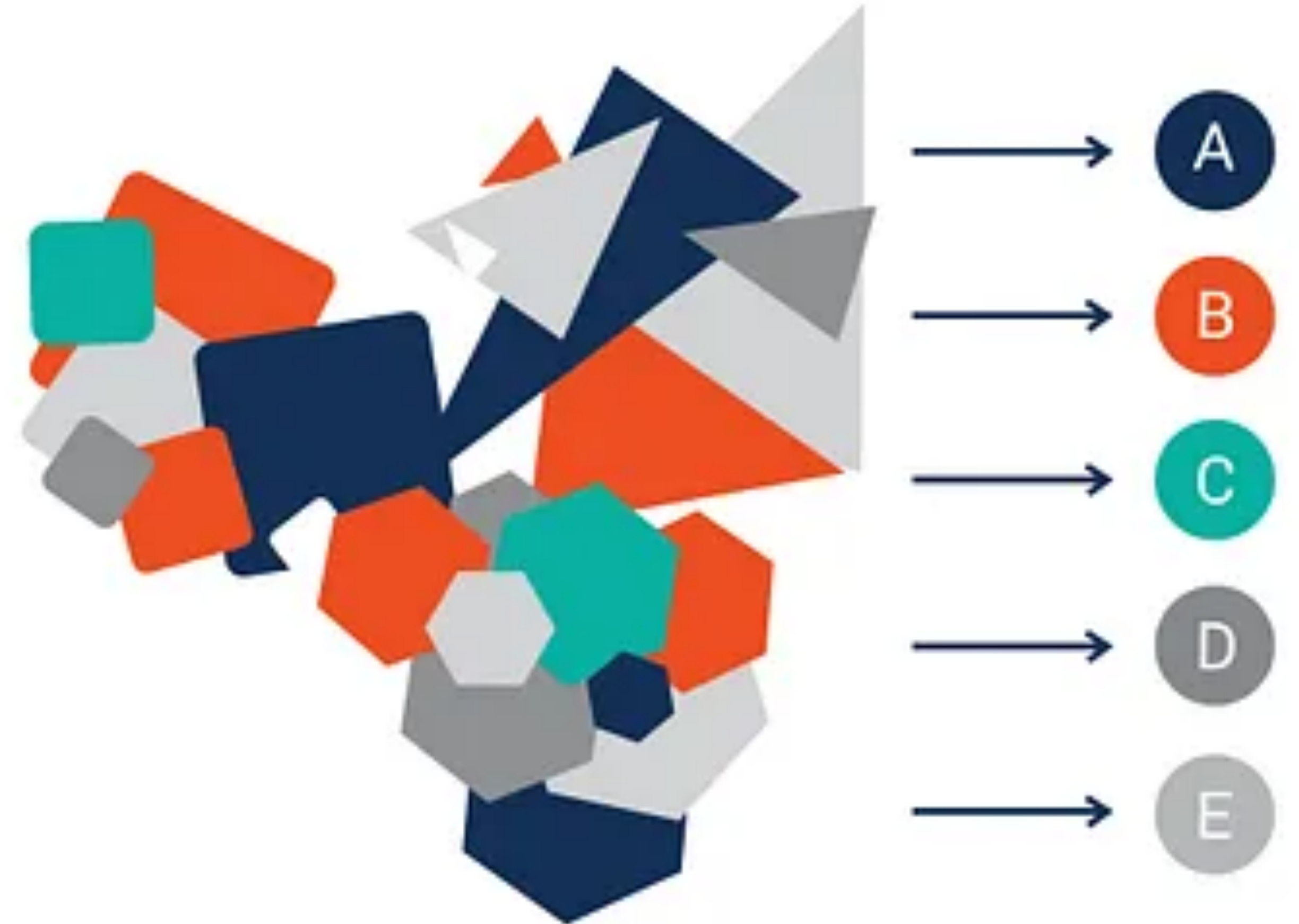
1. Information collected

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Paper vs. Digital forms

Digital forms advantages

Paper

Manual transcription = errors

Delays (weeks/months before data usable)

Data often “stuck” in local offices

Hard to validate or correct in real time

Digital Forms

Enumerators enter directly on device

Immediate sync + backup

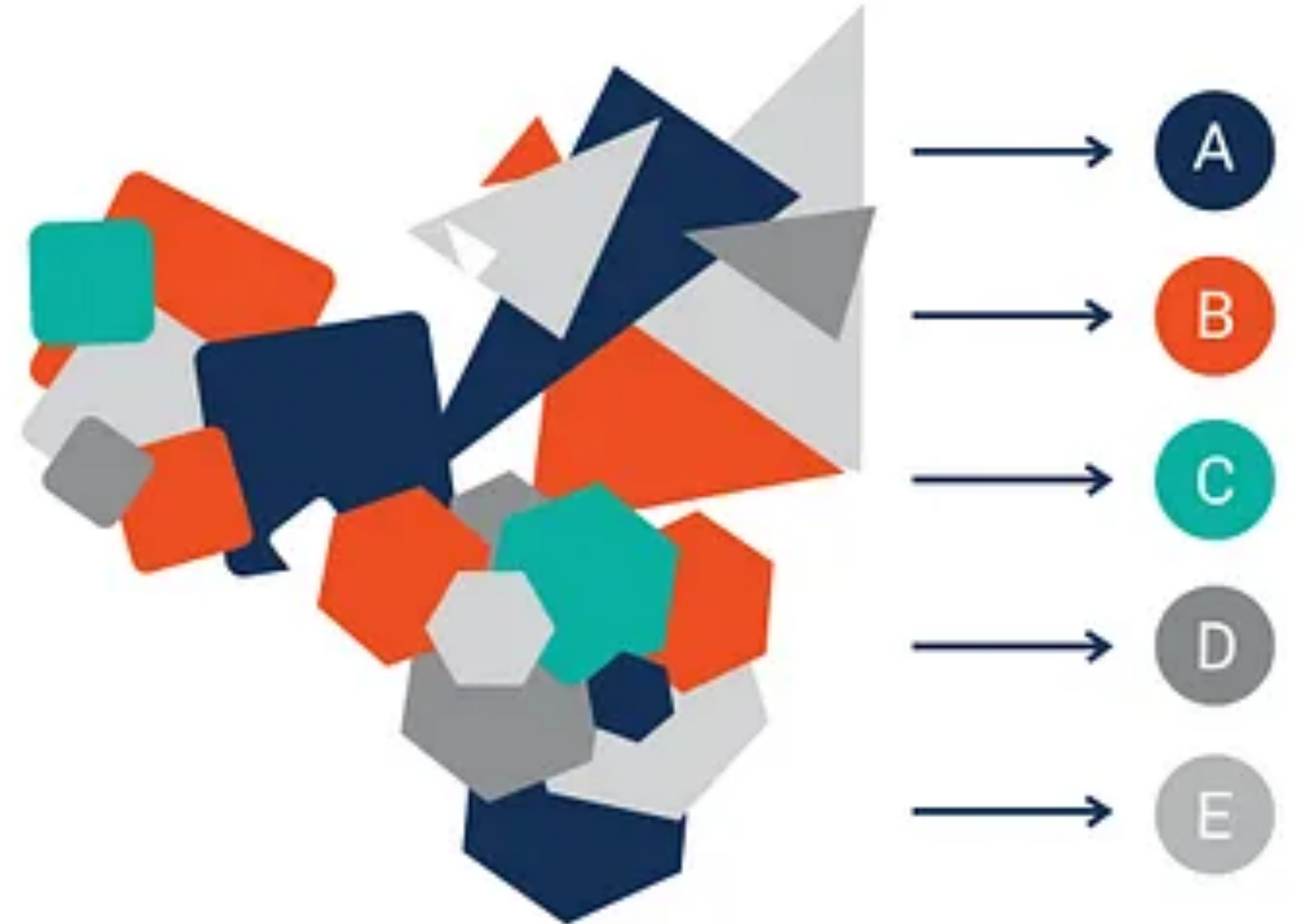
Built-in validation

Automatic export into analysis pipelines

Data collection heterogeneity

One resource, many approaches

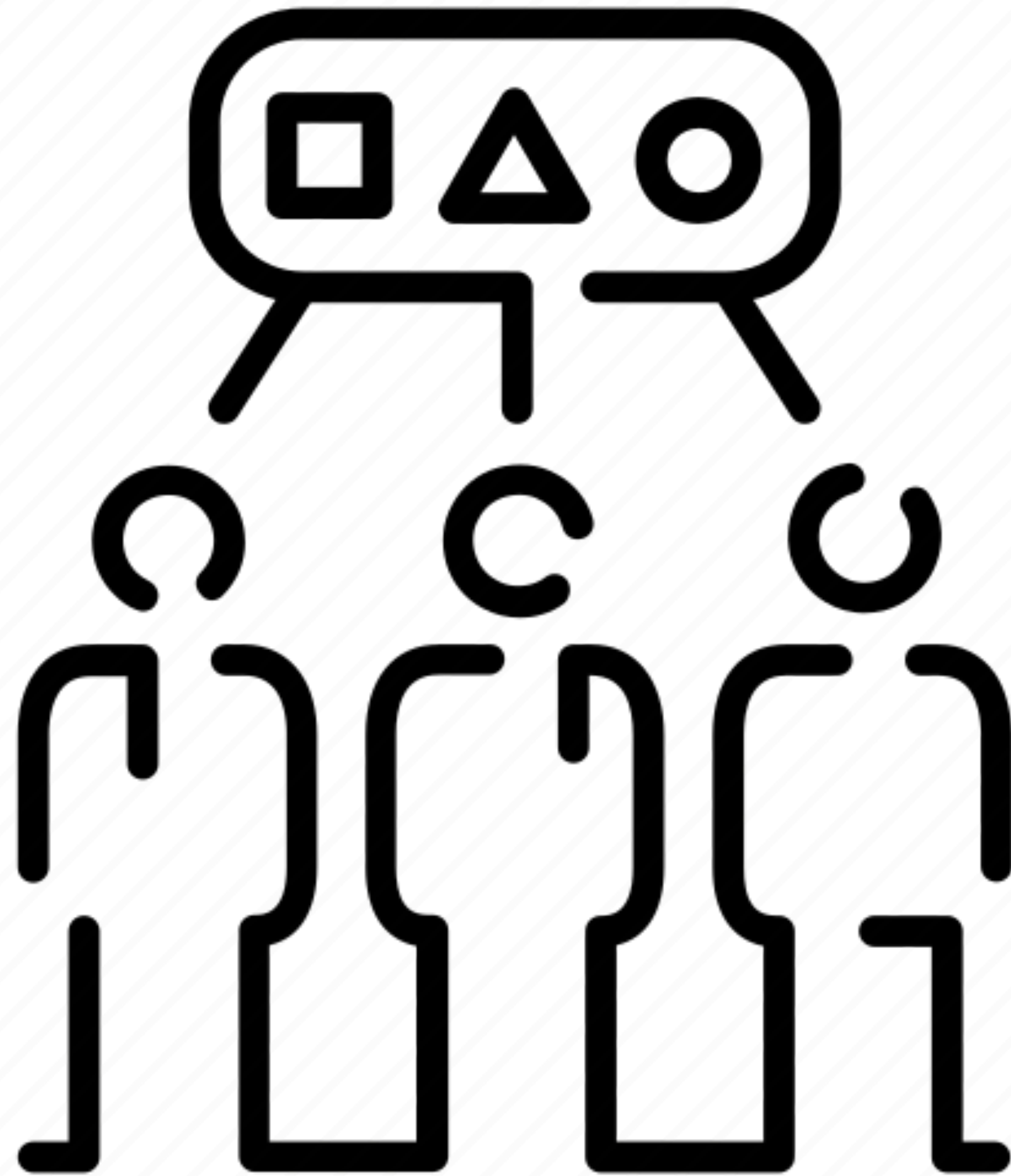
1. Information collected
2. Paper vs. Digital forms
- 3. Terms/variables heterogeneity**
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Terms/variables standardisation

The challenges

Concept	Org. 1	Org. 2	Org. 3	Org. 4
Landing Site / Area	Area	Landing site	Region	Landing site
Boat Type	Boat Type	Vessel type	Type of boat	Vessel
Gear	Gear	Gear	Type of gear	Gear
Effort	N. fishers	Trip duration (Days)	Effort (hours)	Fishing Time



Are we referring to the same thing?

You call it **Monofilament Nets** I call it **Gillnets**

You call it **Linha de mão** I call it **Handlines**

You call it **Pointed Sticks** I call it **Stick Rod**

Your terms are in **Swahili**, mine are in **English**

Terms/variables standardisation

Practical solutions - Classifications and ontologies



Food and Agriculture
Organization of the
United Nations

FAO
FISHERIES AND
AQUACULTURE
TECHNICAL
PAPER

ISSN 2070-7010

672

Classification and illustrated
definition of fishing gears

Gear categories (First tier)	Subcategory (Second tier)	Standard abbreviations	ISSCFG code
GILLNETS AND ENTANGLING NETS			07
	Set gillnets (anchored)	GNS	07.1
	Drift gillnets	GND	07.2
	Encircling gillnets	GNC	07.3
	Fixed gillnets (on stakes)	GNF	07.4
	Trammel nets	GTR	07.5
	Combined gillnets-trammel nets	GTN	07.6
	Gillnets and entangling nets (nei)	GEN	07.9

Ontologies > AQFO

Aquatic Food Ontology (AQFO)

OWL

View license

Watch (0)

Last submission date June 16, 2025

SummaryClassesPropertiesInstancesNotesMappingsWidgetsSPARQL

English

Jump to

> diagnosis

> feed ingredient

> fish feed qualities

> fish treatment

fisheries monitoring

> gear

> fishing net

fuel

> line

manual - by hand

Details

Instances (0)

Visualization

Notes (0)

Mappings (0)

ID

http://w3id.org/aqfo/aqfo_00002220

Preferred name

gear

Synonyms

fishing gear

Definitions

A fishing gear is a tool or method used to catch fish, such as hook-and-line, trawl net, gillnet, pot, trap, spear, manual collection etc.

Raw data

Terms/variables standardisation

Practical solutions - Classifications and ontologies

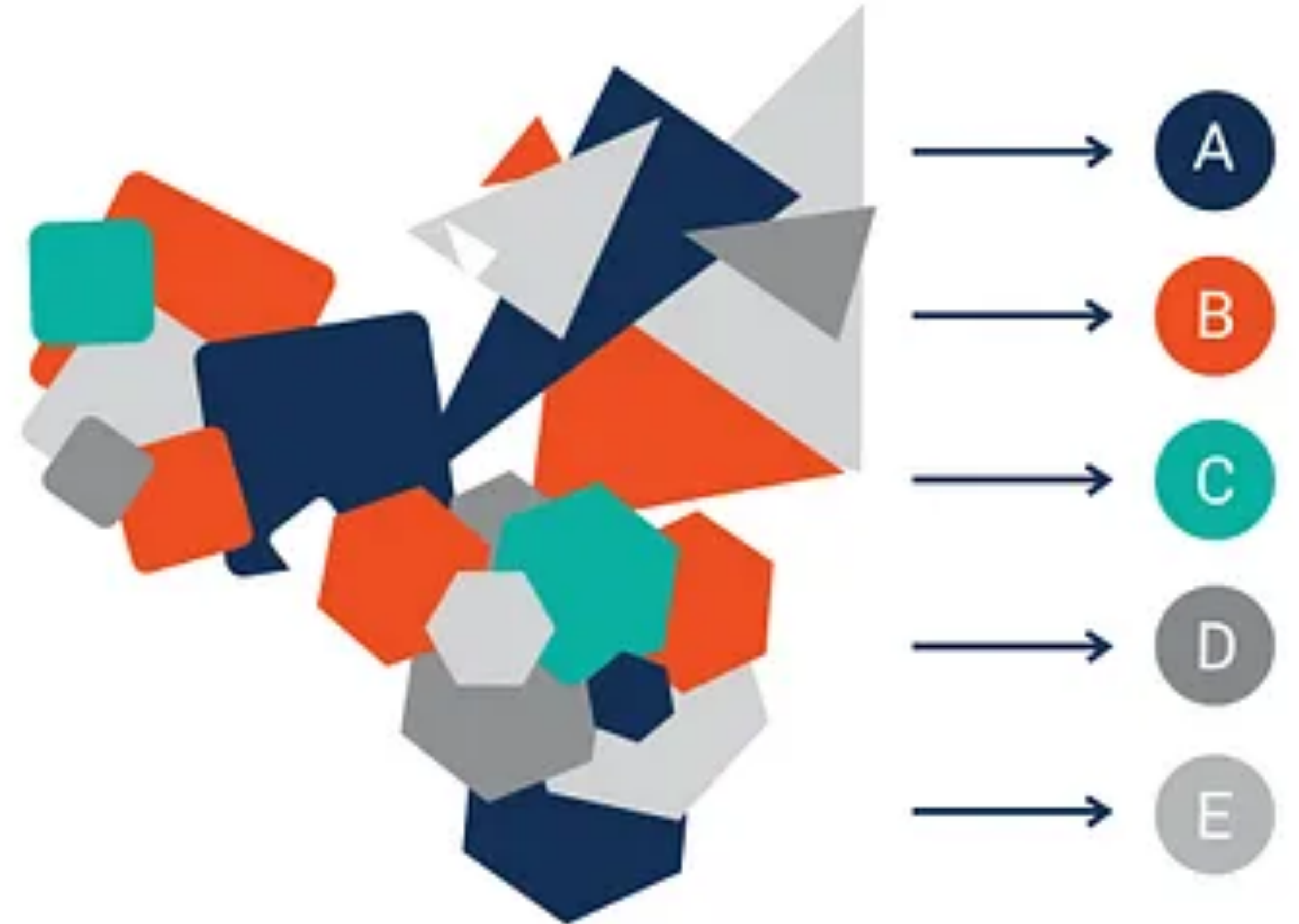


Ensures each variable name has a single, clear definition and the various collections speak the “same language”

Data collection heterogeneity

One resource, many approaches

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Different Species Lists

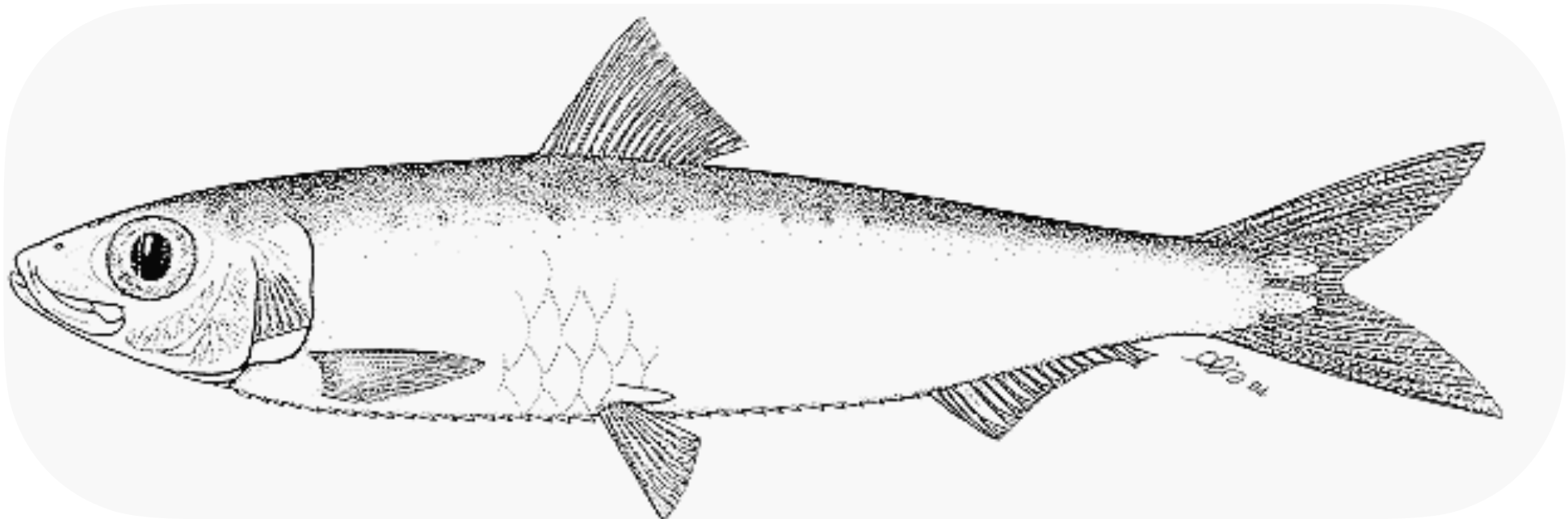
The challenges

Organization	Taxonomic Resolution
Org. 1	Species-level
Org. 2	Mixed Resolution (species/genus)
Org. 3	Species-level (species/genus)
Org. 4	Mixed Resolution

Enumerators training is crucial

Linguistic barriers

Sardina pilchardus



Hello, my name is:

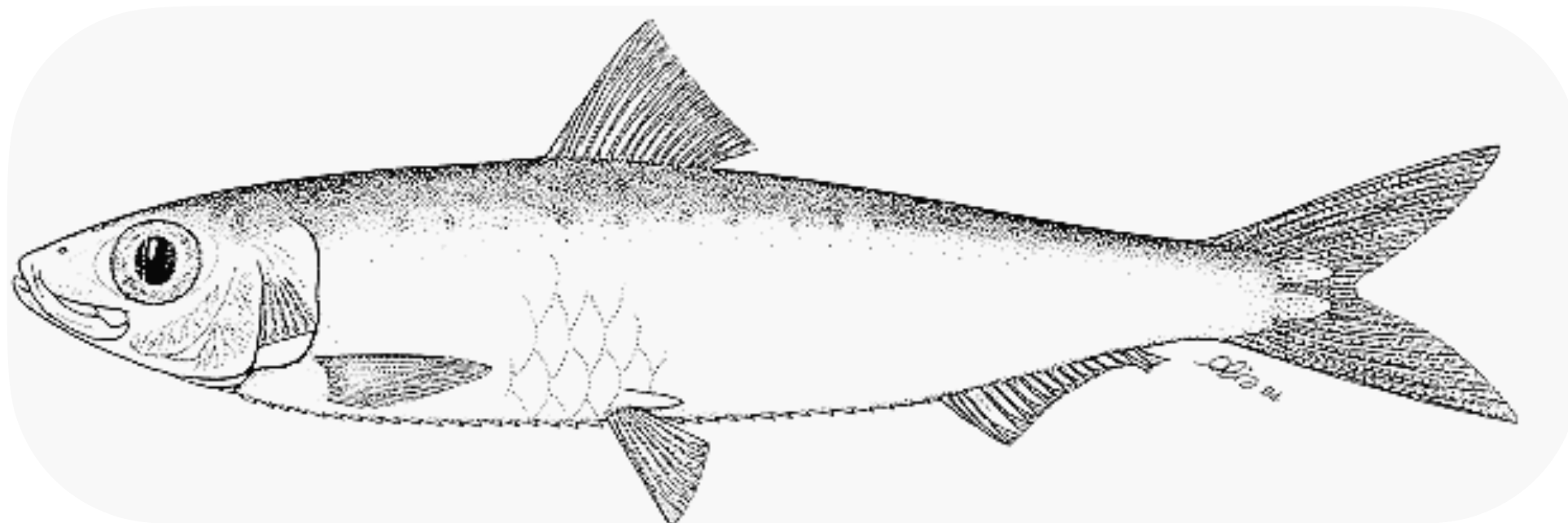
- Pilchard
- Sardina
- Sardinopa-chilena
- Sardella
- Sardine
- Sardinha

Different Species Lists

Practical solutions

- Adopt a reference guide that pairs different organizations local names, scientific names to FAO 3-alpha codes
- Common species photo guide shared among the organizations
- Encourage species photo submission

Sardina pilchardus



Hello, my name is:

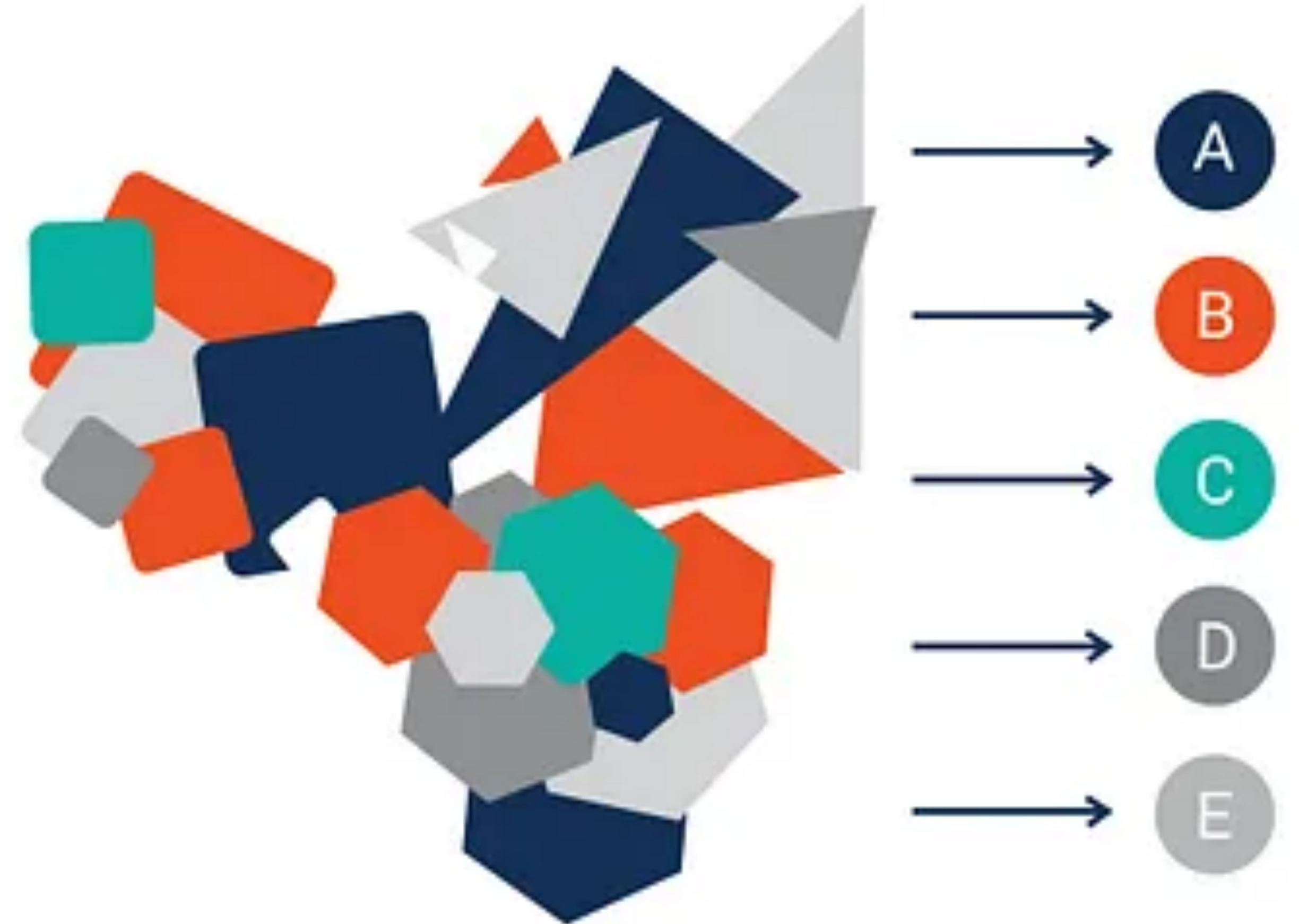
PIL (FAO 3-alpha code)

Data collection heterogeneity

One resource, many approaches

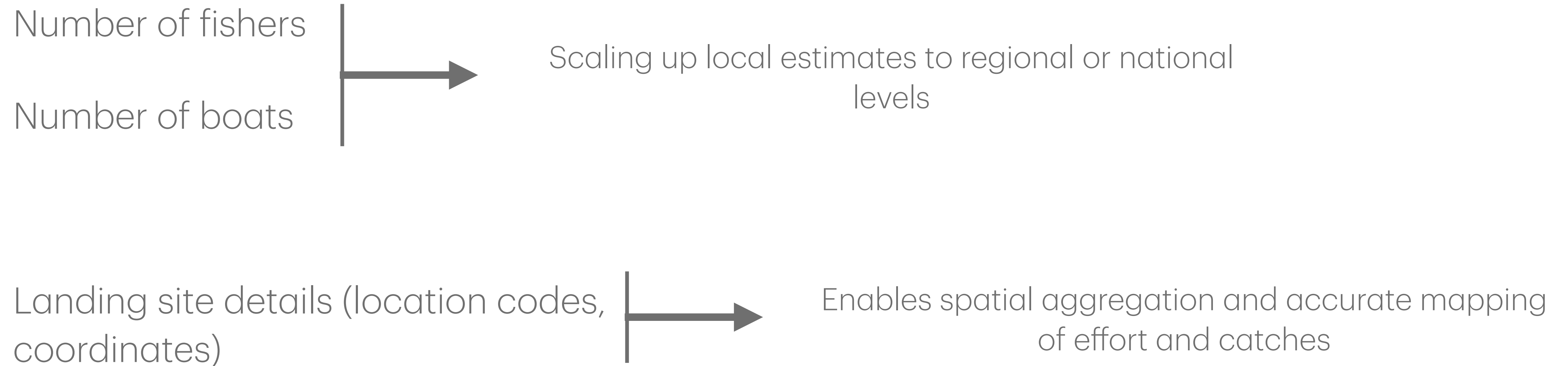
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Why Fishery Information Matters

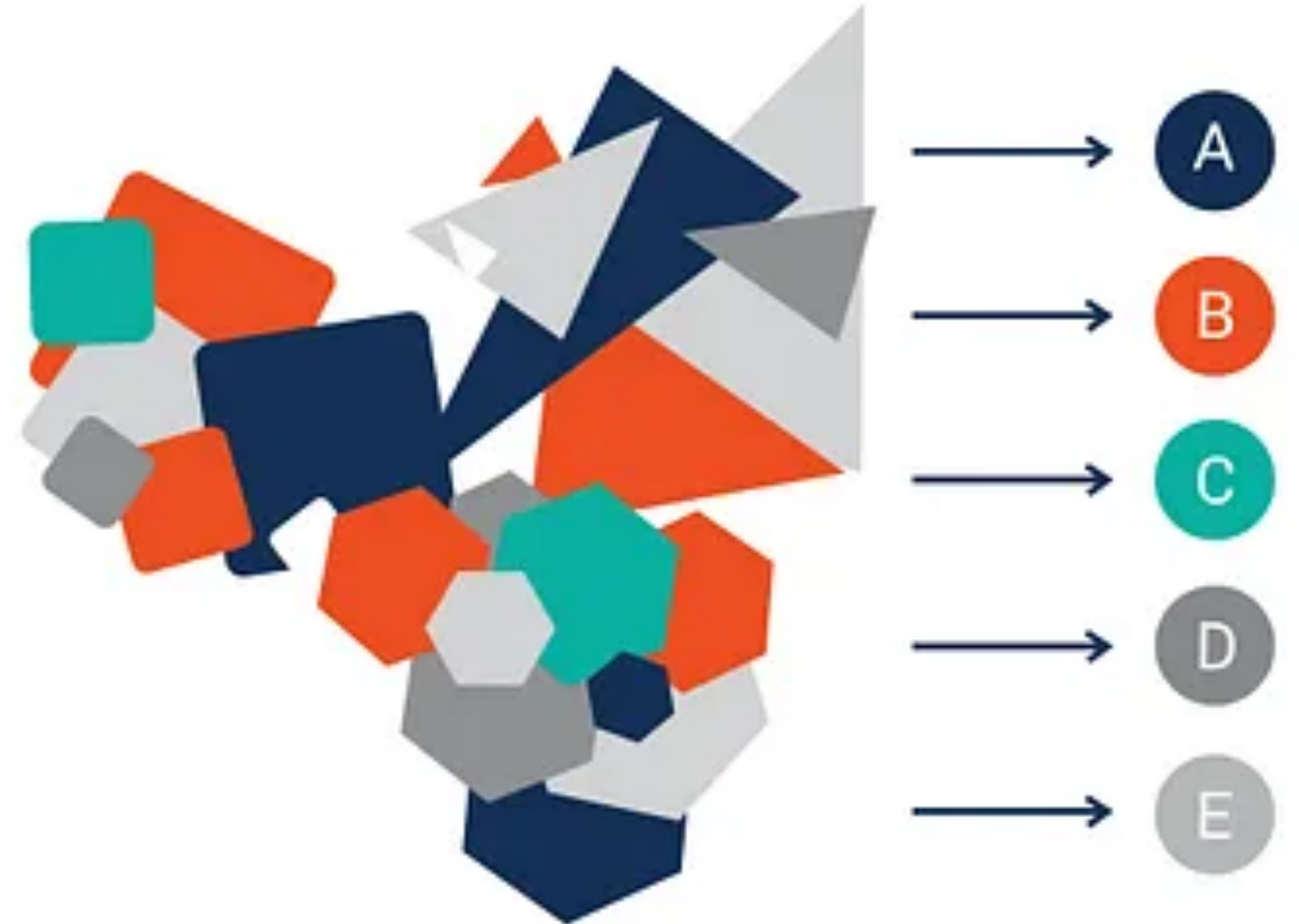
Scale up local datasets to regional level



Data collection heterogeneity

One resource, many approaches

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Practical actions

- **Implement digital workflows**

Transition from paper to digital forms (e.g., KoboToolbox)

- **Adopt a shared data dictionary**

Align variable names with a recognized ontology

Align to classifications systems and ontologies for gear types, vessel types, landing sites, habitats, etc.

- **Species verification**

Align taxonomic units to FAO standard references

Unique photo guide among all enumerators

- **Align key fishery metrics**

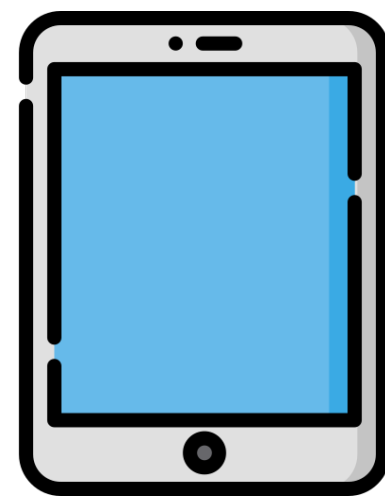
Ensure forms capture comparable effort metrics to inter-district comparison and regional scaling

Create or update a shared fishery registry (boats, fishers, landing sites) to scale local data to regional or national levels

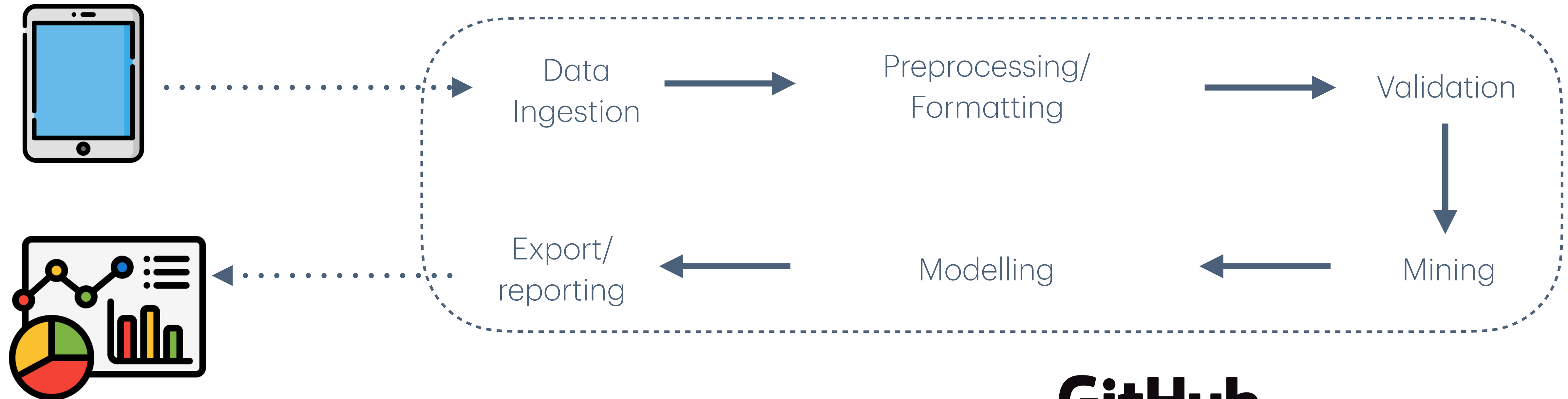
PESKAS

A case study

Enumerators collect data in the field through tablets



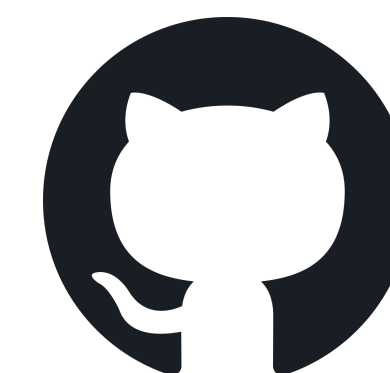
PESKAS data modules



Stakeholders get information through reports and dashboards

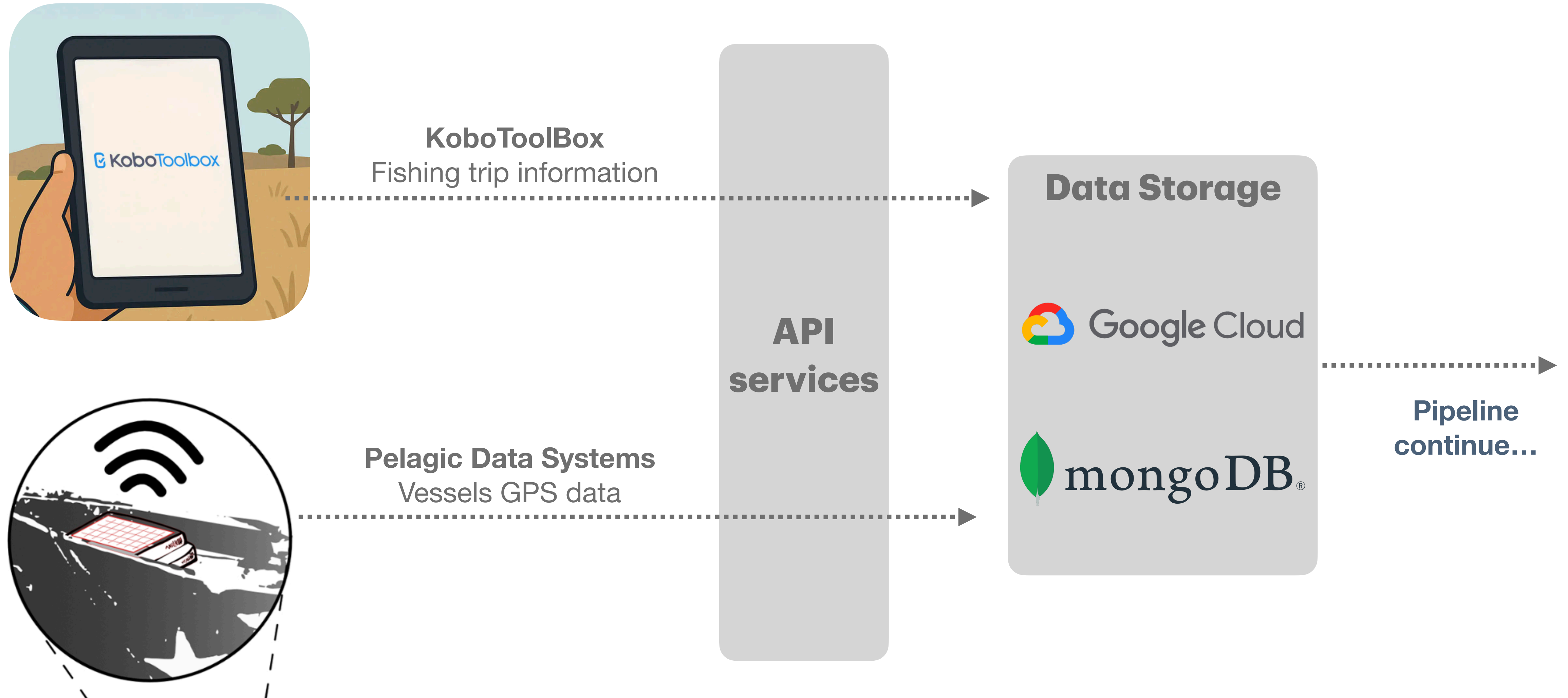
Automated daily workflow

GitHub



Data Ingestion

Getting data from multiple sources (running everyday)



Data Preprocessing

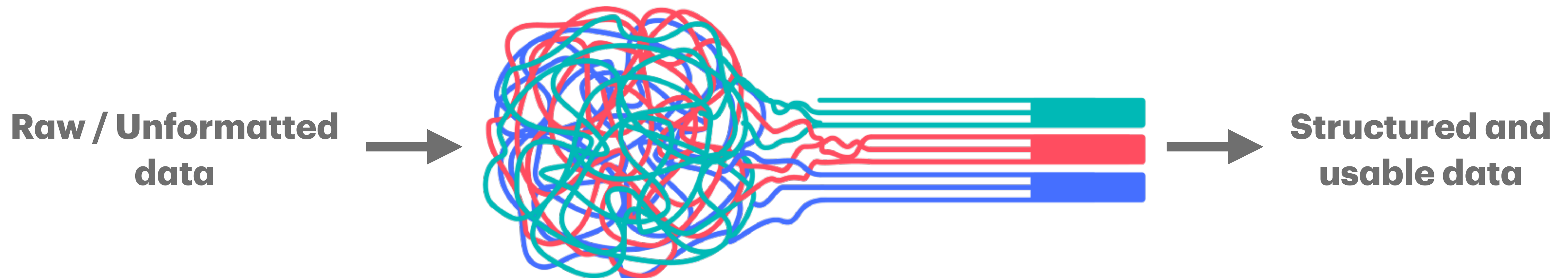
Make data usable and comparable

Data Cleaning & Formatting

- Removes redundant information and raw data artefacts
- Standardizes data formats (tidy format, apply recognised variable names conventions)
- Assigns appropriate fields to data (use ISO 8601 format for dates etc...)

Terminology Standardization

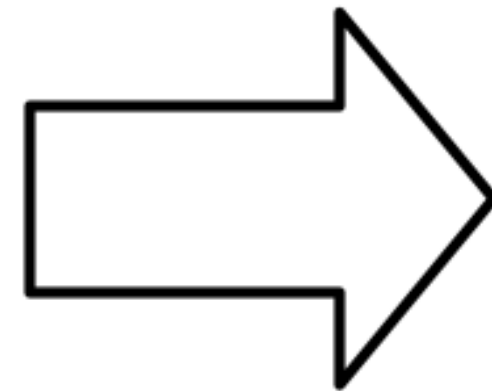
- Aligns with Aquatic Foods Ontology / FAO classifications
- Consistent naming across variables
- Facilitates crossregion and crosscountry comparison



Data format

The importance of the data “shape”

id	bp1	bp2
A	100	120
B	140	115
C	120	125



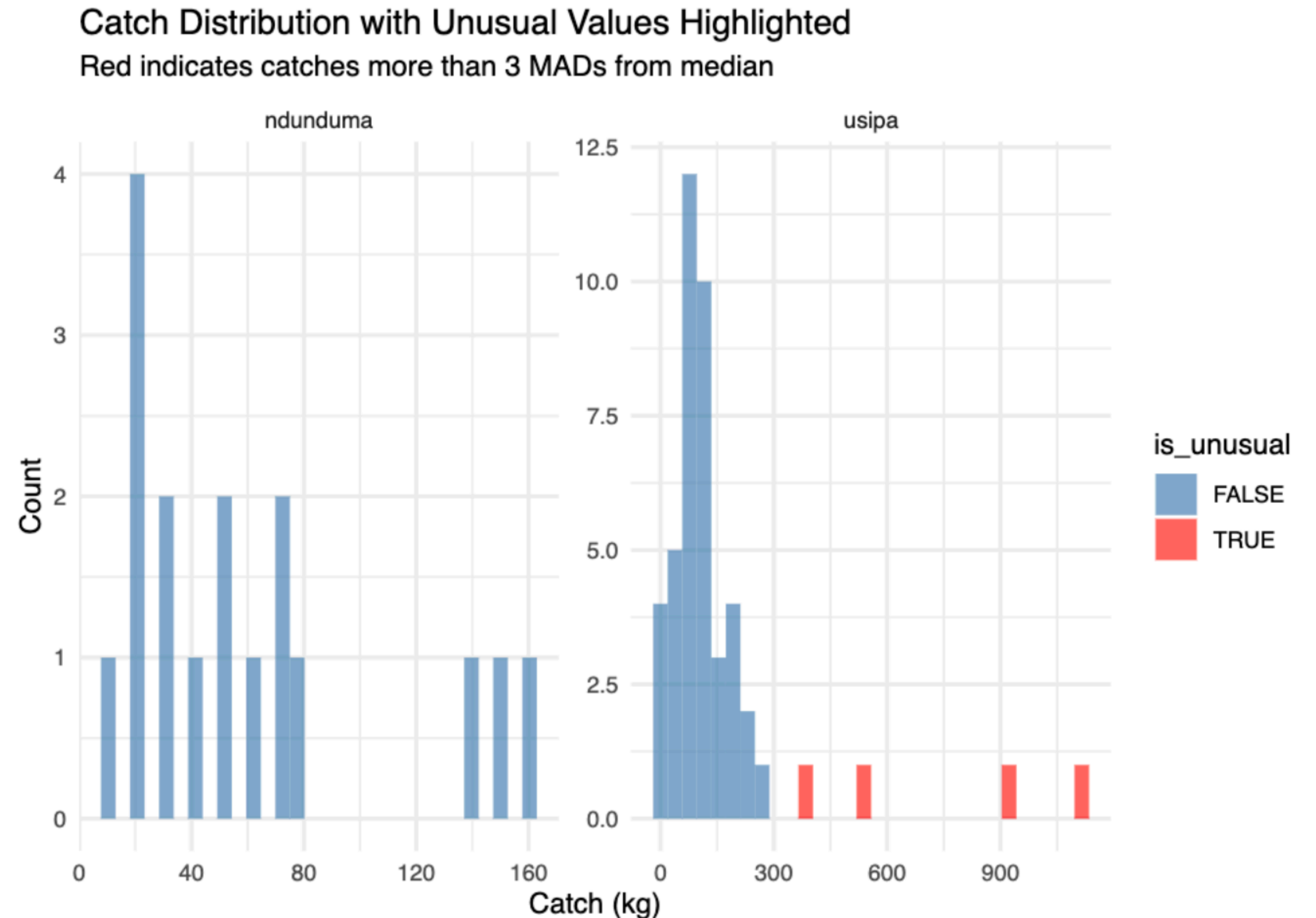
id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
B	bp2	115
C	bp1	120
C	bp2	125

Data Validation

Automated Outlier Detection

Data come with human errors:

- Overestimated/underestimated fish parameters
- Overestimated/underestimated revenue
- Anomalous fishing trip information
- Identifies anomalous values using statistical methods
- Employs both univariate (MAD) and multivariate approaches
- Automatic reporting of enumerators performance



Data Validation

Validation Portal

 Search all columns...

StatusAll Statuses

AlertAll Items

From19/02/2025To02/10/2025

Reset Filters

SUBMISSION ID ↓	SUBMITTED BY ↓	SUBMISSION DATE ↓	ALERT ↓	STATUS ↓	LAST VALIDATED ↓
645323011	wfzbr11	2025-02-19	—	Approved	02/10/2025, 15:13
645323528	wfzbr7	2025-02-19	—	Approved	02/10/2025, 15:14
645326567	wfzbr18	2025-02-19	—	Approved	02/10/2025, 15:14
645327348	wfzbr9	2025-02-19	5, 2	Not Approved	02/10/2025, 14:57
645327358	wfzbr7	2025-02-19	—	Approved	02/10/2025, 15:14
645327970	wfzbr11	2025-02-19	—	Approved	02/10/2025, 15:14
645331603	wfzbr18	2025-02-19	—	Approved	02/10/2025, 15:14
645337833	wfzbr18	2025-02-19	3	Not Approved	02/10/2025, 14:57
645342506	wfzbr12	2025-02-19	—	Approved	02/10/2025, 15:14
645345830	wfzbr10	2025-02-19	5	Not Approved	02/10/2025, 14:57

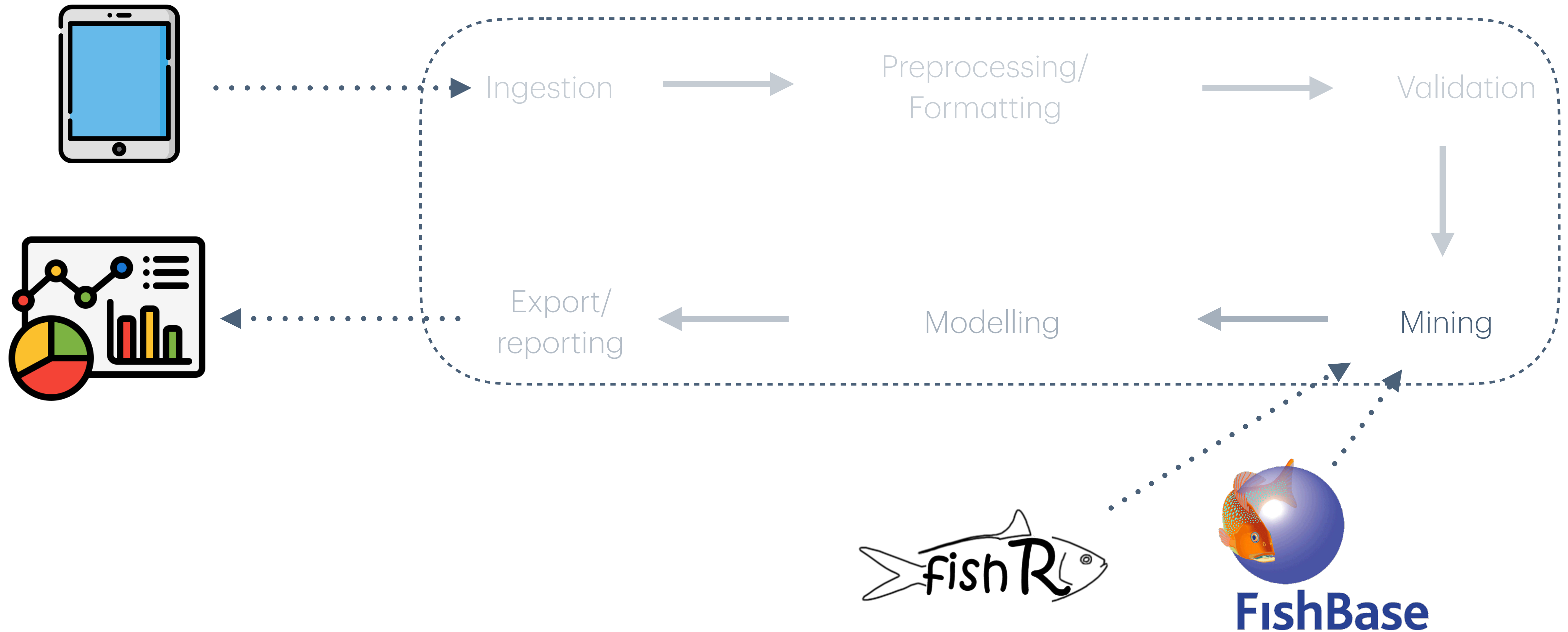
Show 10

Page 1 of 467 4668 results

PreviousNext

PESKAS approach

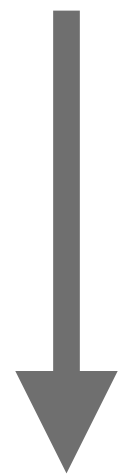
Data Mining



Data Modelling

Scaling to regional level

- Vessels activity coefficient
- N. Boats by district
- Catch per fishing trip

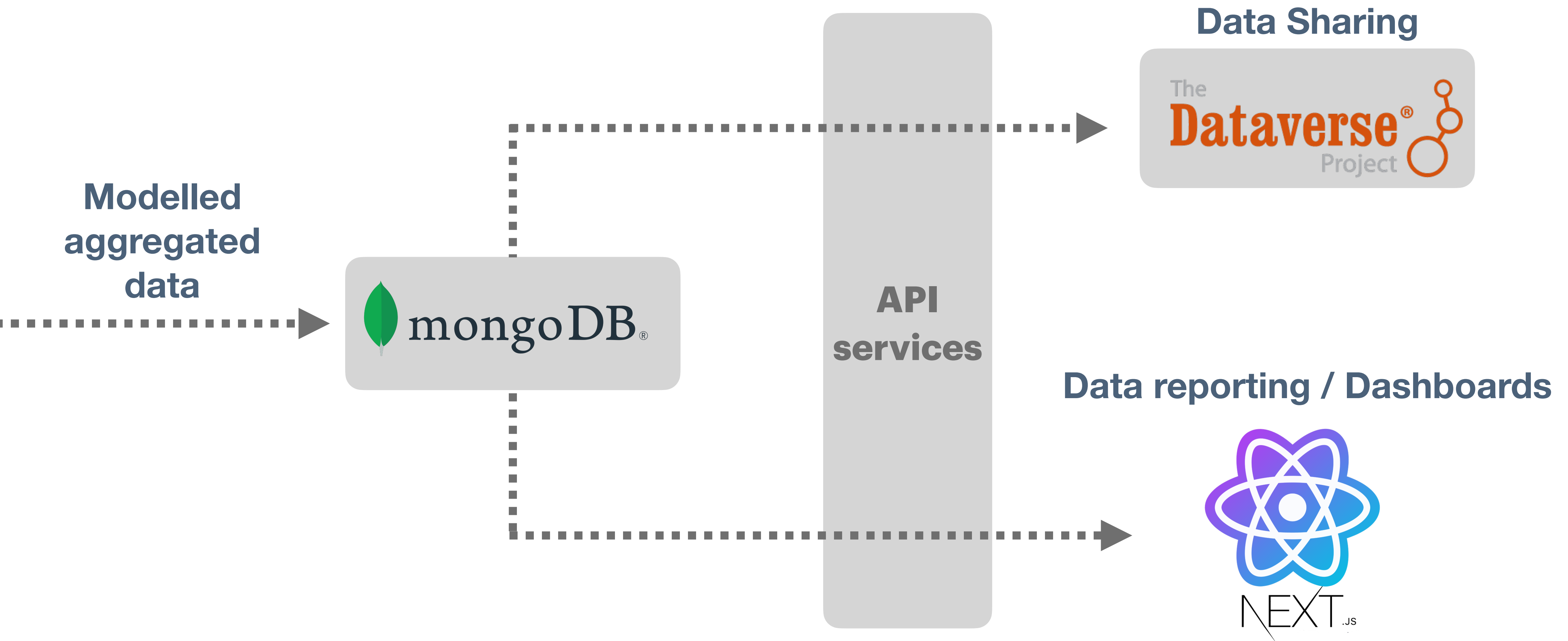


Districts / National estimates



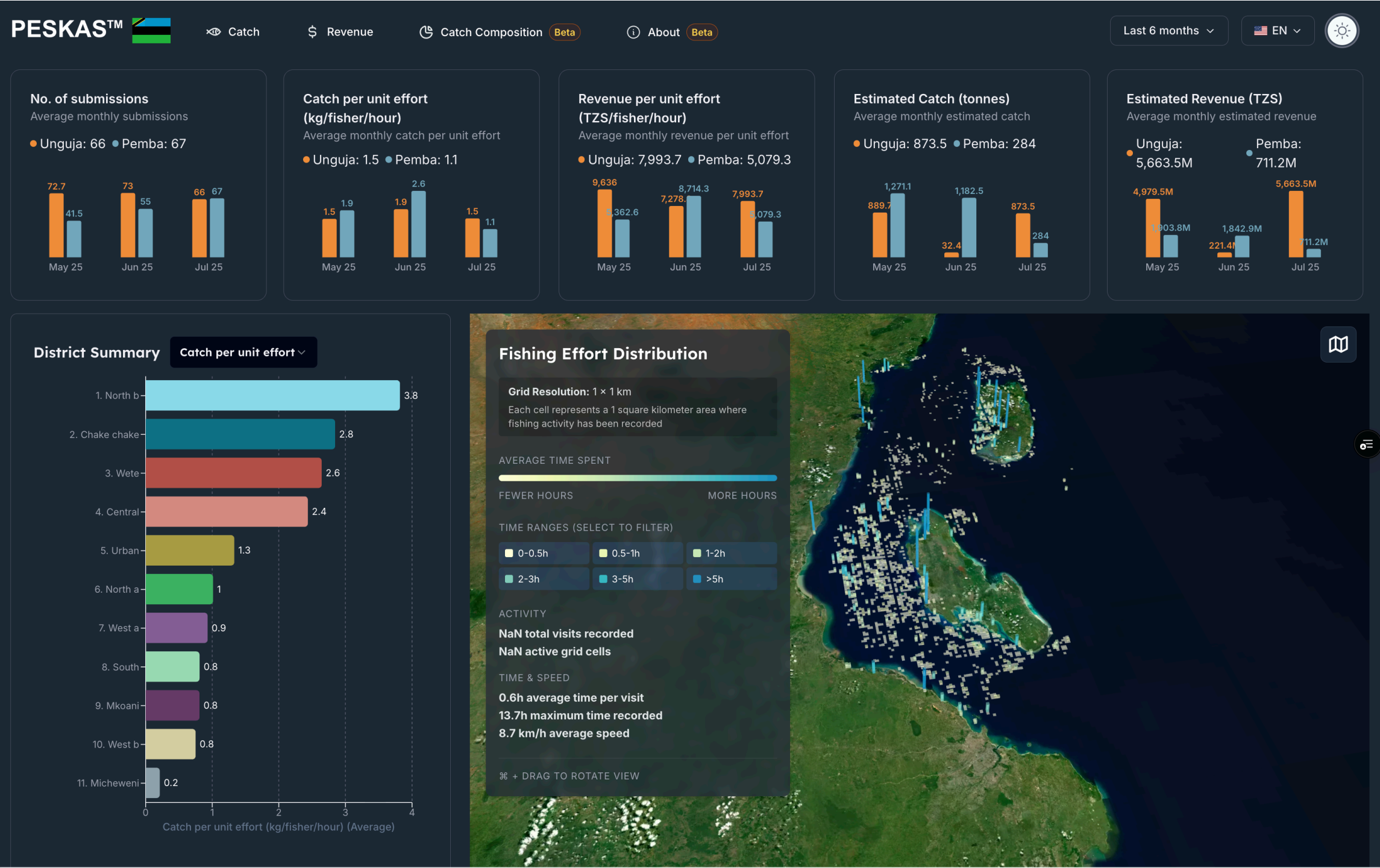
Data Export

Sharing and Reporting

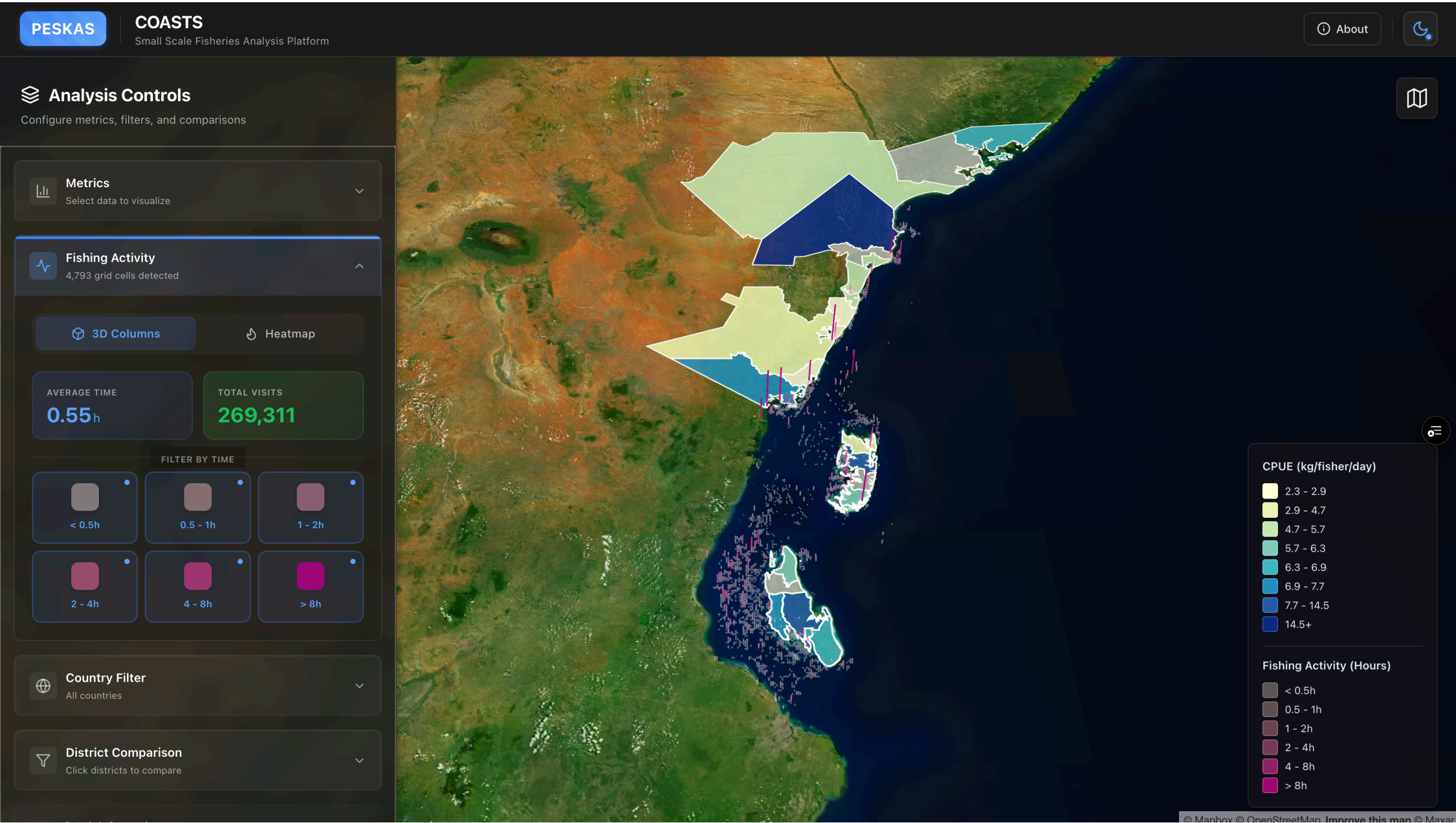


Some examples

PESKAS Zanzibar



COASTS



Thank you!



↕
Data
storage



- TimorLeste
- Kenya
- Zanzibar
- Mozambique

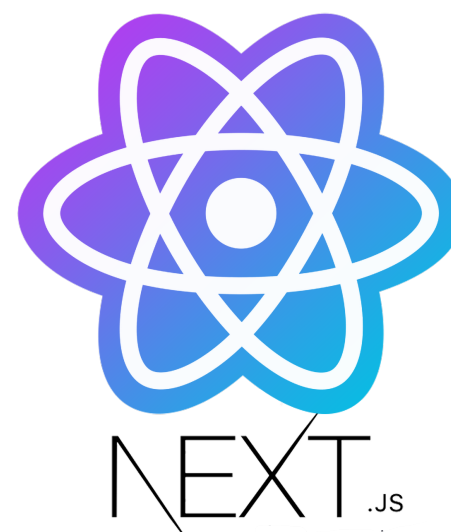
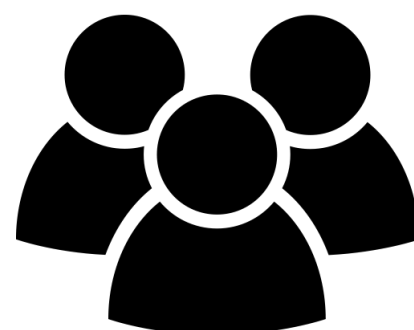
GitHub

- Version Control
- Continuous integration
- Automated workflow with **GitHub Actions**

- R package
- Modular code



Community / Stakeholders
feedback



Data sharing

Peskas web portal application

Dynamic data report